

*****UNCLASSIFIED*****



Replace Chiller York 2075

Prepared by:
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160 Main Road, Building 16
Yorktown, Virginia 23691-1009

Submitted/Prepared by:
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Date: 1/30/2026

Approved by:
Requirements Branch Head:
Bruce Rausch

1. POINTS OF CONTACT:

FEAD Contracting Officer: Maurice Muse, (757) 887-4705
FEAD Contract Specialist in charge: Kiana Owens, (757) 887-4195
Facility Maintenance Specialist (FMS): Brent Walters, (804) 815-3659
Customer POC:

2. GENERAL:

- 2.1 The contractor shall provide supervision, labor, material and equipment required to perform all work for the following project: **Replace Chiller at York 2075**. Report any unforeseen findings to the attention of the Contracting Officer Representative (COR) assigned to the project. The COR is the individual designated as the authorized representative of the Contracting Officer. The COR is responsible for monitoring the performance and technical management of the effort required and should be contacted regarding questions or problems of a technical nature. The COR will be the construction manager (CM), engineering technician (ET), and/or performance assessment representative (PAR) assigned to the project. Materials mentioned herein shall be the same or equal to specified manufacturer and or in quality.
- 2.2 All work shall be done in a manner to provide a complete, usable, and finished product. See paragraph "3. WORK REQUIREMENTS" for an expanded and detailed scope of work.
- 2.3 Contractor shall provide multiple quotes (preferably 3 for each Division); if unable to provide multiple quotes include documentation. In the absence of multiple quotes, a complete breakdown of prices is required for each trade or subcontractor quote to support or justify their price and establish adequate competition.
- 2.4 A pre-proposal site meeting shall be held at the project site to review and discuss the scope of work and assess existing field conditions, prior to award of this task order. The contractor and other key personnel involved shall attend the site meeting with the Government. The contractor shall coordinate/schedule the date and time with the government COR.
- 2.5 Existing reference drawings may be available upon request. Reference drawings shall be used for informational purposes only and may not accurately reflect current as-built conditions. The contractor is responsible for verifying all measurements at the jobsite before ordering materials. All quantities are approximations and shall be verified by the contractor.
- 2.6 All work shall be performed in accordance with Unified Facilities Criteria (UFC's), standard industry practices, local, state and federal codes, laws and regulations.
- 2.7 The contractor shall maintain an organized, clean storage area on-site and will be responsible for securing material in a manner as to not impede the work of Government personnel. The jobsite will be left clean daily, and all construction debris properly disposed of off the installation at the contractor's expense in accordance with all local, state, federal, and regulations to include NAVFAC MIDLANT environmental compliance mandates.
- 2.8 The contractor shall request a final acceptance inspection for project closeout with the COR. Any punch list shall be corrected prior to final payments.

3. WORK REQUIREMENTS:

3.1 SOW:

This document outlines the scope of work for the replacement of a TRANE 100-ton chiller unit (Serial #U13L39763, Asset #713354) and the relocation of the existing unit on Yorktown Naval Weapons Station. The project involves the decommissioning of the existing 100-ton chiller, its relocation to a new site approximately two (2) miles away near the chiller unit at York 1595, and the complete supply, installation, and commissioning of a new high-efficiency 100-ton air-cooled scroll packaged chiller at the original location. Replace all insulation,

heat tape, aluminum jacketing, pressure and temperature gauges on water lines above existing concrete pad. The contractor is responsible for providing all labor, materials, tools, equipment, and supervision required to execute this project safely and in compliance with all applicable codes and standards. November through April is preferred for construction.

3.2 Contractor Responsibilities:

The contractor shall be responsible for, but not limited to, the following:

- Furnishing a new chiller unit and all related materials and equipment as specified in Section 3.0.
- Providing all necessary project management, labor, tools, and supervision.
- Submitting a detailed NAVFAC Crane Lift Plan in compliance with Army Corps of Engineers Safety Manual EM-385, for review and approval prior to any lifting operations.
- Submitting a comprehensive Electrical Safety Plan, detailing procedures for lockout/tagout (LOTO) and the safe connection of the new electrical service.
- Ensuring all work is performed in a workmanlike manner and adheres to all applicable federal, state, and local codes, regulations, and safety standards.
- Obtaining all required permits for the project.
- Coordinating all utility shutdowns (electrical, water) required for demolition and installation.
- Ensuring the safe removal, transportation, and re-installation of the existing chiller unit.
- Providing comprehensive testing, balancing, and commissioning of the new system.
- Providing training for on-site maintenance staff.
- Restoring all work areas to their original or better condition upon project completion.

3.3 New Chiller Specifications

The new unit provided must meet or exceed the following technical specifications:

- Chiller type: Air-Cooled Scroll Packaged Chiller
- Nominal capacity: 100 tons
- Electrical: 208 Volt / 3 Phase / 60 Hertz
- Power rating: Max. Current (MCA): 447.0 Amps /Max. Overcurrent Protection (MOP): 600 Amps
- Performance: High efficiency
- Refrigerant: R-454B
- Protection: Factory-installed flow freeze protection
- Valves: Refrigerant isolation valves (including discharge valve)
- Flow switch: Factory-installed
- Pipe connection: Grooved pipe connection
- Insulation: Factory-installed 0.75" insulation
- Fins: Wide ambient temperature, lanced aluminum fins
- Starter: Across-the-line / Direct-on-line (DOL) starter
- Power connection: Single-point connection for main unit power
- Control panel: High-fault rated control panel with circuit breaker
- Short Circuit Rating: High Ampere (A) short circuit rating
- Enclosure: UL 60335 rated for outdoor applications
- Control interface: BACnet MS/TP interface
- Vibration isolators: Elastomeric isolators (field-installed)
- Strainer: Factory-installed water strainer
- Aesthetics: Architectural louvered panels

3.4 Decommissioning and relocation of existing chiller

- Perform a safe and controlled shutdown of the existing 100-ton chiller unit
- Evacuate and reclaim all refrigerant in accordance with all EPA and local regulations
- Disconnect all electrical power, control wiring, and water piping
- Execute the approved crane lift plan to safely rig and lift the chiller from foundation
- Secure the chiller onto appropriate transportation

- Transport the unit to York 1595 (approximately 2 miles) NOTE: transport and placement only (not connection)
- Offload and set the existing chiller at the relocation site

3.5 Installation procedure for new chiller unit

- Inspect the existing structural supports and concrete pad. Report any deficiencies.
- Replace all existing water pressure and temperature gauges (4) on the chilled water lines with new calibrated gauges
- Install new insulation on all exposed chilled water lines, ensuring complete coverage and vapor sealing
- Connect the new single-point main power to the unit in accordance with the approved Electrical Safety Plan and the National Electric Code (NEC) to include conduit.
- Replace flexible conduit coupler on both service lines
- Connect the BACnet MS/TP interface to the building's automation system (BAS)

3.6 Startup Procedure and Acceptance Testing

- Perform a full system startup and pre-functional checklist according to the manufacturer's specifications
- Verify and record operating voltages and amperages at full load conditions
- Test and balance the chilled water flow to meet design GPM'
- Run the chiller for a minimum of four continuous hours under building load to verify stable operation
- Demonstrate successful integration and communication of all specified points via the BACnet MS/TP interface to the BAS
- Log all performance data (refrigerant pressures, water temperatures, amperage) during testing for inclusion in the final report

3.7 Project Submittals/Deliverables: See item #7 below

4. SPECIAL SCHEDULING REQUIREMENTS/WORK PROCEDURES:

- 4.1 Work Hours:** The contractor's normal working hours shall be 0630 to 1700, Monday – Friday excluding federal holidays. The contractor must submit in writing any requests to work outside the normal working hours and only the contracting officer can approve or deny these requests. All personnel must be authorized and vetted to work on site.
- 4.2 Passes and Badges:** All contractors, subcontractors and vendors shall comply with the installation security requirements regarding personnel, vehicle, equipment, security passes/badges and access to the jobsite. All contractors shall have valid installation access passes and badge to perform work on an awarded contract.
- 4.3 Access to Buildings/Occupied Buildings:** Do not enter any facilities without prior approval from the assigned COR.

5. SAFETY:

- 5.1** All Safety and Hot Work contact information and procedures will be discussed at the pre-con meeting. For emergencies on WPNSTA Yorktown call Fire and Rescue (757) 887-4911.

Hot Work Permits OUTSIDE of the Restricted Area – Industrial Areas / Yorktown / Cheatham Annex Base:

- 1) All welding and burning operations, or large grinding and cutting operations, require a Yorktown Hot Work permit and the Fire Marshal will be called and notified by the GOV representative (PAR, ET, CM, FM or other) and or the KTR of the work location, type of work and dates for the work to be performed.
- 2) The Fire Marshal shall inspect and advise for the Hot Work conditions and limitations and the Hot Work Permit will be issued for the as needed time requirements to complete the work.

Hot Work Permits INSIDE of the Restricted Area - Yorktown:

- 1) All welding, burning, and grinding operations that create arc or spark conditions require a Hot Work Permit and notification to the Yorktown base Fire Marshal, that the work area and facility is within the Restricted Area, with work conditions and locations as stated above.
- 2) This also includes gas engine electric generators, gas engine pumps, refueling of gas / diesel fuel operated equipment and standalone refueling tanks.
- 3) The Fire Marshal will notify the Yorktown Explosive Safety office and both the Fire Marshal, and the Explosive Safety inspector will review the work location and site for approval of the Hot Work Permit.

The base Fire Marshal for Yorktown phone # is (757) 613-8194 and for Cheatham Annex is (757) 508-0076

- 5.2 All contract work shall be accomplished IAW manufacturer's recommendations, industry standards, Unified Facilities Criteria (UFC), Unified Facilities Guide Specifications Section 01 35 26, State and local governing authority, EM 385-1-1, Safety and Health Requirements Manual (current edition) and Occupational Safety and Health Administration requirements. If there is conflicting guidance the most stringent specifications shall apply.
- 5.3 The contractor shall submit an initial Activity Hazard Analysis (AHA) to identify actual or potential hazards associated with specific work activities to be performed and an Accident Prevention Plan(s) (APP) for preventative measures the contractor will take before beginning any work task, activity, or defined feature of work (DFOV). The AHAs shall define the steps being performed within the activity, task or DFOV and identify the work sequences. Specific anticipated hazards, site conditions, equipment, materials, personnel, and the control measures to be implemented; it shall also address life safety measures and personnel egress. The contractor must obtain government approval prior to starting any construction activities.
- 5.4 EM 385-1-1 dated March 2024 "Safety and Health Requirements Manual
- 5.5 29 CFR 1910 (OSHA) "2015" General Industry Regulations"
- 5.6 **EXCAVATIONS** - Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1

UTILITY LOCATIONS - Provide a third party, independent, private utility locating company to positively identify underground utilities in the work area in addition to any station locating service and coordinated with the station utility department.

UTILITY LOCATION VERIFICATION - Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within 3 feet of the underground system.

UTILITIES WITHIN OR UNDER IMPERVIOUS SURFACES - Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with station utility departments in addition to location and depth verification by a third party, independent, private locating company. The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

- 5.7 Lockout/tagout (LOTO) practices shall follow EM 385-1-1 and 29 CFR 1926.417 Electrical Circuits, and UFGS 01 35 29 and may only be performed by authorized employees. LOTO tags shall have standardized print and format, constructed and printed to prevent UV light, wet locations, or corrosive environment will not deteriorate the tag, attach in a reusable method and warn against hazardous conditions.

6. HAZARDOUS MATERIALS DISCLOSURE STATEMENT AND ENVIRONMENTAL COMPLIANCE:

- 6.1** If a contractor suspects hazardous materials or substances are present, he or she must stop work immediately, notify the COR and wait for further guidance.
- 6.2** For projects involving less than 10,000 square feet of land disturbance or when no land disturbance is proposed but has the potential for storm water pollution still present i.e. painting, service connections, minor building renovations, hazardous material abatement, etc. The contractor must utilize best management practices (BMP) to ensure materials associated with this activity do NOT contaminate storm water runoff i.e. paints, solvents, petroleum products, trash, sediment, etc. The Contractor shall keep a spill kit on hand in case of any equipment failure, hydraulic line break, etc. if heavy equipment will be used for this project. No discharges to the sanitary sewer system, storm system or surface waters are permitted without prior coordination and approval from the Water Program Manager. Concrete mixing and/or pouring must be done in a way to prevent entry to storm water conveyances or surface waters. Notify NAVFAC environmental office immediately if there are any spills or contaminants entering the storm water or sewer systems. *Regulatory authorities must be notified within 24 hours of the discovery. All waste disposal is required to be in accordance with federal, state, and local regulations and laws.
- 6.3** If temporary above ground storage tanks (AST) or containers that have a storage capacity of 55 gallons or greater, secondary containments are required along with spill control equipment and are subject to inspections. The tank program manager is required to be contacted for assistance and guidance for temporary storage of Petroleum Oils Lubricants (POL) on the project site. For spills i.e. petroleum, hazardous substances etc. contact the tank program manager for assistance.
- 6.4** If land disturbance for this project exceeds one acre the following provisions shall apply: Erosion and Sediment Control Plan (E&S), Stormwater Pollution Prevention Plan (SWPPP), and Stormwater Management Plan (SWMP) must be developed and approved in accordance with the Virginia E&S Control Law and Regulations, VSMP Law and regulations. The E&S Control Plan must be developed in accordance with the Virginia Erosion and Sediment Control Handbook, latest edition; the SWMP must be developed in accordance with Section II of the VSMP Construction General Permit. A copy of the approved E&S Control Plan, SWM Plan and SWPPP must be retained on-site and complied with throughout project execution. Since E&S Controls are necessary for this project the operator should be prepared to budget for the necessary installation and maintenance of E&S control measures to include expenses for possible additional controls, if necessary, throughout the duration of the project.
- 6.5** If during ground disturbing activities archaeological resources are discovered, work in the immediate area of the find must stop and cultural resources manager needs to be contacted immediately. Please note if the project design or scope of work changes, the cultural resources manager needs to be contacted immediately as State Historic Preservation Office (SHPO) consultation may be required at that time.
- 6.6** NEPA: A documented NEPA decision i.e. signed Record of Categorical Exclusion (RCE) must be completed prior to undertaking any action or award of any contract. Failure to follow the requirements subject the installation to unnecessary risk for environmental violations, repeat or additional regulatory consultations/permits, schedule, and cost overruns.
- 6.7** Wastewater (CWA/HRSD) POC: Jason Oliver (757) 887-4808 or Jason.j.oliver@navy.mil
Environmental Compliance POC: Hannah Jeff (757) 353-8158 or hannah.r.jeff.civ@us.navy.mil
Cultural Resources Program Manager POC: Tiffany Goodrich (757) 887-4756 or tiffany.goodrich@navy.mil
For spills (i.e. petroleum, hazardous substances, etc.) contact: NWSY/CAX Regional Dispatch Center at (757) 887-4911
- 6.8** Work which requires the disturbance or removal of any of the following: mercury and LLR components and PCBs shall be performed in accordance with 29 CFR 1910.1000, 10 CFR 20, 40 CFR 273, and 40 CFR 761.

7. SUBMITTALS/DELIVERABLES:

- 7.1 The contractor shall submit the following for government acceptance prior to the start of any construction activities:
- 7.2 Approved Site Safety and Health Officer (SSHO), employee must work for the prime contractor.
- 7.3 Approved Quality Control employee (must be employed by the prime contractor)
- 7.4 Approved Site Superintendent (must also be employed by the prime contractor)
- 7.5 Approved project schedule
- 7.6 Accident Prevention Plan (APP), must be site specific with Activity Hazard Analysis (AHA) for each Definable Feature of Work
- 7.7 Emergency Action Plan, can be separate or an attachment to Accident Prevention Plan
- 7.8 Quality Control Plan, will identify the Contractor's Organizational Chart and provide associated documentation, methods and testing to ensure quality
- 7.9 Material/Equipment Submittal Plan will identify projected dates that material submittals shall be forwarded to the government. Contractor must provide proof of manufacturing. Purchase from the vendor to the contracting officer (KO) and request a no-cost time extension via the Request for Information (RFI) process
- 7.10 Material/Equipment Laydown Area Plan shall identify areas where Contractor plans to store equipment and materials during construction and is subject to approval by COR and PW Environmental Division
- 7.11 Schedule of prices: Include a detailed breakdown of the contract price with quantities for each kind of work. Include general conditions, profit and overhead in the unit prices. Break down into design and each category if stated in attachment. The contractor may invoice for bonds once the Government has approved the bonds, however, no other request for payment will be processed without an approved Schedule of Prices.

8. SUBMITTAL REGISTER:

- 8.1 Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements. Contractor must check and approve all items prior to submittal and stamp, sign and date indicating action taken. Proposed deviations from the contract requirements shall be clearly identified. Include within submittals items such as: Contractor, manufacturer, or fabricator's drawings; descriptive literature including but not limited to, catalog cuts, diagrams, operating charts or curves, test reports, test cylinders, samples, Operations and Maintenance (O&M) Manuals (including parts list), certifications, warranties, and other such required submittals.
- 8.2 Project schedule
- 8.3 Accident Prevention Plan (must be site specific)
- 8.4 Activity Hazard Analysis
- 8.5 Emergency Action Plan (OSHA 29 CFR 1910.38)
- 8.6 Quality Control Plan

- 8.7 Material/Equipment Submittals i.e. manufacturer's product data, color and product samples, cut sheets, installation and/or maintenance instructions (O&M) and warranty
- 8.8 Environmental Plan i.e. permitting, controls, waste management etc.
- 8.9 Proof of manufacturing purchase from vendor as requested
- 8.10 Safety Data Sheet (SDS) as applicable for products used on site
- 8.11 Licenses and permits
- 8.12 Digging permit (14 days prior to excavation)
- 8.13 NAVFAC Red-Zone (NRZ) Checklist/POAM Closeout submittals UFGS-01- 78 00
- 8.14 Warranty Management Plan (including warranty letters and POC information) Five (5) year manufacturer's warranty for the new chiller unit. A comprehensive one (1) year warranty covering all labor and workmanship.

9. ATTACHMENTS:

- 9.1 Site Location Map & Photos

*****End of SCOPE OF WORK*****